LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principle of operation and constructional details of float type gauge
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principle of operation and constructional details of float type gauge. Practical & Demonstration
5. Specific Objectives	15.1.7.2 Trainee will be able to explain float type fuel quantity measurement indicator.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of float type gauge mechanism using model; Safety: Handle float arm gently to avoid damage; Supervision: Instructor monitors trainee interaction with model components
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a fuel quantity indicator?
(b) Name two types of fuel gauges used in aircraft.
(c) How does the float type gauge differ from the capacitance type gauge?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Constructional details of float type gauge • Float arm and lever mechanism • Magnetic reed switch operation • Fuel tank installation requirements • Electrical circuit connections	L & Ds	PPT, IB	15 min
(b)	• Principle of operation explanation • Float movement and switch actuation • Signal transmission to cockpit indicator • Calibration and zeroing procedures • Common failure modes and causes	L & Ds	PPT, PDF	20 min
(c)	• Practical demonstration of gauge operation • Verification of switch actuation points • Troubleshooting simulated faults • Documentation of observations • Safety checks during practical session	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the float movement translates into an electrical signal.
(b) Describe the role of the magnetic reed switch in the gauge operation.
(c) List three potential causes for inaccurate fuel readings in a float type gauge.

12. Home Assignment and Follow Up	Prepare a brief note comparing the float type gauge with the capacitance type gauge, focusing on their operating principles and applications, to be discussed in Lesson Plan 18.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Capacitance type fuel gauge.	Refer to the master syllabus reference list	LP 18

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min